



**Faculty of Institute of Science and Technology
Kathmandu, Nepal**

**Nepathya College
Tilottama, Rupandehi, Nepal**

**A CASE STUDY ON
“THE NAGARIK APP ECOSYSTEM”**

For

E-Governance (CSC 337)

A Case Study Submitted in Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Computer Science and Information Technology (B.Sc. CSIT)

**Under the Supervision of
Madav Kahar Goud**

**Submitted By :
Adarasha Gaihre**

Roll no: 2

Email: aakku@adarashagaihre.com.np

1. Background Study -----	1
1.1 Overview of E-Governance	1
1.2 Origin and Development of the Nagarik App	1
1.3 The Nagarik App Ecosystem	1
1.4 Core Characteristics	2
1.5 Relevance to E-Governance in Nepal	2
2. Problem Statement -----	2
2.1 Limitations of Conventional Service Delivery ...	2
2.2 Limitations within the Current Nagarik App Ecosystem	3
3. Possible Solutions -----	4
3.1 Legal and Policy Harmonisation	4
3.2 Verification-Based Interoperability	4
3.3 Agency Onboarding and Capacity Building	4
3.4 Assisted and Omnichannel Access	4
3.5 Comparison of Solution Approaches	4
4. Most Implementable Method in the Context of Nepal --	5
4.1 Why Verification-Led Interoperability Should Come First	5
4.2 Why Legal Harmonisation Must Proceed in Parallel	5
4.3 Why Assisted Access Should Be Retained	6
4.4 A Recommended Phased Roadmap	6
5. Challenges to Implementation -----	6
5.3 Security and Privacy of Centralised Identity Data	7
5.4 Digital Divide and Accessibility	7
5.5 System Reliability and User Experience	7
5.6 Sustainability, Funding, and Capacity	7
6. Suggestions and Feedback -----	7
6.1 For DoIT and Service Agencies	7
6.2 For Policy Makers	7
6.3 For Banks, Telecom Operators, and Other Institutions	8
7. Conclusion -----	9

Abstract

This case study examines the Nagarik App ecosystem, the Government of Nepal's flagship platform for delivering public services and digital documents through a single window. It begins by outlining the concept of e-governance and tracing the development of the Nagarik App, including its integration with the National Identity system. It then identifies the structural problem faced by the platform: although the technology supports real-time verification, banks, hospitals, and government offices continue to demand physical documents owing to legal and institutional constraints. Four solution approaches are outlined, namely legal harmonisation, verification-based interoperability, agency onboarding, and assisted access. The study concludes that a phased, verification-led approach, supported by parallel legal reform and continued assisted access, is the most implementable path. Key challenges and practical recommendations for agencies, policy makers, and institutions are also presented.

1. Background Study

1.1 Overview of E-Governance

E-governance is the use of information and communication technology by government to deliver services, exchange information, and engage citizens in an efficient, transparent, and accountable manner. It is commonly classified into Government-to-Citizen (G2C), Government-to-Business (G2B), and Government-to-Government (G2G) interactions, and typically rests on three foundations: a trusted digital identity, interoperable agency systems, and a single point of access for the citizen.

1.2 Origin and Development of the Nagarik App

The Nagarik App is the Government of Nepal's flagship citizen services application. It is developed by the Department of Information Technology (DoIT) under the Ministry of Communication and Information Technology (MoCIT), with the National Information Technology Center (NITC) supporting the national digital infrastructure. It was introduced in 2021 as a component of the Digital Nepal Framework, initially in beta, with the objective of consolidating services from multiple government agencies into a single mobile platform.

1.3 The Nagarik App Ecosystem

The Nagarik App is best understood as an ecosystem rather than a standalone application. Its principal layers are as follows:

- **Citizen interface:** the Nagarik mobile application and web access point used by citizens.
- **Agency systems:** services and documents contributed by individual agencies, such as the Inland Revenue Department (PAN), the Election Commission (voter details), Nepal Police (clearance reports), and transport authorities (driving licence).
- **Identity layer:** the National Identity (NID) system, integrated into the application in January 2025, which enabled identity-linked services such as passport applications and account opening.
- **Service catalogue:** by early 2025, more than 60 services from roughly 35 government entities were reported to be available through the application, with further services, including No-Objection Certificates, Public Service Commission admit cards, and professional certificates, being added in 2026.

1.4 Core Characteristics

- a single window for services across multiple government agencies.
- identity-linked access through registered mobile number, OTP verification, and NID details.
- digital versions of official documents, such as citizenship, driving licence, PAN, and voter details.
- federated ownership, in which DoIT operates the platform while individual agencies own their respective service components.
- round-the-clock availability, with services added incrementally over time.

1.5 Relevance to E-Governance in Nepal

Nepal's public administration has historically relied on physical visits, paper records, and repeated submission of the same documents to different offices. Rising smartphone and internet access, together with the introduction of the National ID, has made an identity-centred digital service platform both feasible and necessary. The Nagarik App therefore represents a practical and timely case for examining how e-governance can be implemented, and where it encounters constraints, in the Nepali context.

2. Problem Statement

The Nagarik App has expanded steadily in scope, yet its value depends on whether citizens, agencies, and private institutions accept digital records as equivalent to physical ones. Both the conventional service model and the current digital model present significant limitations.

2.1 Limitations of Conventional Service Delivery

- Citizens must visit multiple offices in person, incurring cost and time, often during restricted office hours.
- The same personal information and supporting documents are submitted repeatedly to different agencies.
- Agency records are fragmented, which limits verification, traceability, and transparency.
- Service standards vary between offices, with limited visibility into application status.

2.2 Limitations within the Current Nagarik App Ecosystem

- Banks, hospitals, and government offices continue to request physical documents despite the availability of verified digital versions, forcing citizens to carry both.
- Several older Acts require physical documentation, and DoIT lacks the authority to compel other institutions to accept digital records; a regulation cannot override an Act.
- Service components belong to different agencies, resulting in uneven availability, quality, and update speed across services.
- Digital access and literacy remain unevenly distributed between urban and rural populations.
- Centralised storage and linkage of identity data raises concerns regarding security, privacy, and platform reliability.

2.3 The Core Problem

The central problem addressed by this case study is: how can the Nagarik App ecosystem evolve from a repository of digital documents into a legally recognised, interoperable, and trusted service platform, while operating within the constraints of Nepal's existing legal framework, institutional readiness, digital inclusion levels, and data security requirements.

3. Possible Solutions

No single measure resolves the limitations identified above. Four broad solution approaches can be distinguished, each addressing a different layer of the problem.

3.1 Legal and Policy Harmonisation

Existing Acts and regulations are reviewed and amended so that verified digital documents carry legal equivalence to physical originals. This addresses the root cause of non-acceptance and gives DoIT and partner agencies a clear mandate.

3.2 Verification-Based Interoperability

Institutions are enabled to verify a citizen's identity or document in real time through QR codes or secure APIs linked to the source agency, rather than requesting paper copies. This reduces institutional risk and effort, thereby encouraging voluntary adoption.

3.3 Agency Onboarding and Capacity Building

Additional agencies are onboarded to the platform under common technical standards and service-level targets, supported by training of agency staff. This improves consistency and expands the range of services available.

3.4 Assisted and Omnichannel Access

Citizens who lack devices, connectivity, or digital skills are supported through ward-level and municipal help desks, alongside web access and conventional counters. This ensures that digitisation does not exclude any segment of the population.

3.5 Comparison of Solution Approaches

Approach	Best suited for	Key requirement
Legal and policy harmonisation	Removing the statutory barriers that require physical documents	Amendment of Acts and regulations, with inter-ministerial coordination
Verification-based interoperability	Institutions that need to confirm identity or documents quickly and reliably	Secure QR or API verification, and clear consent rules
Agency onboarding and capacity building	Expanding service coverage and improving service quality	Common integration standards, service-level targets, trained staff
Assisted and omnichannel access	Citizens with limited digital literacy or connectivity	Ward and municipal help desks, and web and offline support

4. Most Implementable Method in the Context of Nepal

Considering the institutional mandate of DoIT, the existing technical capability of the platform, and the shared ownership of services across agencies, a phased, verification-led approach is the most implementable. It prioritises real-time verification for institutions, supported by parallel legal harmonisation, gradual agency onboarding, and continued assisted access for citizens.

4.1 Why Verification-Led Interoperability Should Come First

Officials have indicated that the primary barrier to acceptance is not technology but legal and institutional readiness, as the platform already supports digital verification across agencies. Enabling institutions to verify documents directly delivers immediate benefit without waiting for comprehensive legal reform, and builds the confidence on which wider acceptance depends.

4.2 Why Legal Harmonisation Must Proceed in Parallel

Verification alone cannot compel acceptance where an Act requires a physical document. Targeted amendments, building on existing electronic transaction provisions, are therefore required to secure lasting recognition of digital records. Sequencing these amendments alongside technical rollout avoids delaying visible progress.

4.3 Why Assisted Access Should Be Retained

Digital literacy and connectivity remain uneven. Retaining conventional counters and introducing assisted access at local levels ensures continuity of service and inclusion during the transition.

4.4 A Recommended Phased Roadmap

- 1. Phase 1:** Enable QR and NID-based verification for priority institutions, namely banks, telecom operators, hospitals, and government counters, supported by a formal government directive.
- 2. Phase 2:** Amend relevant Acts and regulations to establish legal equivalence of verified digital documents.
- 3. Phase 3:** Onboard additional agencies through common integration standards and service-level agreements, prioritising high-demand services.
- 4. Phase 4:** Strengthen data protection through consent-based data sharing, access audits, and incident response procedures.
- 5. Phase 5:** Extend ward-level assisted access and additional authentication options, and monitor performance through periodic review.

5. Challenges to Implementation

5.1 Legal and Regulatory Constraints

Provisions in existing Acts continue to require physical documents. Amendment is a lengthy legislative process that depends on coordination among several ministries.

5.2 Institutional Readiness and Coordination

DoIT operates the platform but does not own the service components, and lacks enforcement authority over other agencies. Differences in readiness, priorities, and technical capacity slow integration and produce inconsistent service quality.

5.3 Security and Privacy of Centralised Identity Data

Linking identity, document, and payment-related data increases the potential impact of any breach. Comprehensive data protection legislation and mature security practices are essential to maintain public trust.

5.4 Digital Divide and Accessibility

Limited devices, connectivity, and digital literacy in rural areas restrict the addressable user base. Users abroad or without a registered mobile number may also face authentication difficulties.

5.5 System Reliability and User Experience

Service outages, verification failures, and unclear error handling reduce user confidence and encourage reliance on physical documents.

5.6 Sustainability, Funding, and Capacity

Continued operation, security upgrades, and the addition of new services require sustained funding and skilled technical personnel, which can be difficult to retain within the public sector.

6. Suggestions and Feedback

6.1 For DoIT and Service Agencies

- Publish common integration standards and service-level targets for all agencies onboarded to the platform.
- Provide a simple and well-documented verification facility for institutions, with clear guidance for frontline staff.
- Maintain transparent service status reporting and a responsive grievance mechanism for users.

6.2 For Policy Makers

- Prioritise amendment of Acts that require physical documents where verified digital equivalents exist.
- Enact comprehensive data protection and privacy provisions, with clear accountability for breaches.

- Issue directives requiring public bodies to accept verified digital documents, pending legislative reform.

6.3 For Banks, Telecom Operators, and Other Institutions

- Adopt Nagarik App verification in customer onboarding and service counters, and train staff accordingly.
- Retain physical document alternatives temporarily, while progressively reducing reliance on them.

6.4 General Feedback on the Nagarik App Ecosystem

The technical foundation of the Nagarik App is sound, and its underlying logic of single-window, identity-linked service delivery is well suited to Nepal. The principal gap lies in legal recognition and institutional adoption rather than in the technology itself. Progress should therefore be measured not only by the number of services listed, but by the extent to which digital documents are accepted in practice.

7. Conclusion

The Nagarik App represents a significant step in Nepal's e-governance journey. By consolidating services from many agencies and integrating the National ID, it offers citizens a single, round-the-clock point of access and demonstrates the feasibility of an identity-centred digital service platform.

However, the ecosystem continues to be constrained by non-technical factors: legal provisions requiring physical documents, limited enforcement authority over partner institutions, uneven digital access, and the need for stronger data protection. These constraints, rather than the platform's capabilities, currently determine how much of its potential is realised.

The most implementable path is therefore a phased, verification-led approach, combining real-time verification for institutions with targeted legal reform, gradual agency onboarding, strengthened data security, and continued assisted access. Pursued in this sequence, the Nagarik App can mature from a digital document repository into a trusted, legally recognised backbone of e-governance in Nepal.